

6 Conclusion

We introduced LLM-Prop, a carefully finetuned network derived from T5 for crystal property prediction. We showed, through an extensive set of experiments, that LLM-Prop achieves superior performance in predicting the physical and electronic properties of crystalline solids, outperforming the current state-of-the-art and ubiquitously used GNN-based architectures such as ALIGNN by a large margin on certain tasks. Our results highlight the great potential that text-based methods have in materials science and we release a benchmark text data, TextEdge, composed of text descriptions of crystals and their properties to encourage research in this nascent area.

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